

الجمهورية الجزائرية الديمقراطية الشعبية
وزارة التربية الوطنية

ثانوية بوطبة بشير - المقاومة -
دورة 2024



مديرية التربية لولاية عنابة
امتحان بكالوريا تجريبي للتعليم الثانوي
الشعبة: تقني رياضي

المدة: 04 سا و 30 د

اختبار في مادة الرياضيات

على المترشح أن يختار أحد الموضوعين التاليين

الموضوع الأول

التمرين الأول: (04 نقاط)

(1) $63x + 5y = 159 \dots (E) : (x; y) \quad (E) \quad Z^2$
 $(E) \quad 5 \quad 63$
 $(E) \quad Z^2 \quad x_0 + y_0 = -3 \quad (E) \quad (x_0; y_0)$
 $|13x + y - 33| < 4 \quad (E) \quad (x; y)$
 $5 \quad 3^n \quad n$
 $5 \quad 2023^{2023}$

(2) $35 < n < 65 \quad \begin{cases} 3^n + 3^{4n} - 2024 \equiv 0[5] \\ n \equiv 0[3] \end{cases} : n$
 $5 \quad \overline{\beta 10 \beta 0} \quad 7 \quad \overline{5 \alpha 0 \alpha} \quad A$
 $A + \alpha + \beta \quad \beta \quad \alpha$

التمرين الثاني: (05 نقاط)

(1) $3 \quad 1 \quad 5 \quad 1 \quad 3 \quad 1$
 $"C" \quad "B" \quad "A :$
 7
 $C \quad A \cup B \quad B \quad A$
 X
 $\{0, 1, 2\} \quad X$
 $E(-2X + 3) \quad X \quad E(X)$
 $n \in N^* \quad n$
 $D : p_n$
 $S_n = \sum_{k=1}^{k=n} C_n^k p_k \quad S_n$

$$z_B = \frac{1}{2} + \frac{1}{2}i \quad z_A = \sqrt{3}e^{i\frac{\pi}{6}}$$

$$z_C = -1 + i\sqrt{3}$$

$$z_A$$

$$\frac{3}{2} + \frac{\sqrt{3}}{2}i$$

$$\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

$$\frac{3}{2} - \frac{\sqrt{3}}{2}i$$

$$z_C$$

$$z_C = \sqrt{2}e^{i\frac{2\pi}{3}}$$

$$z_C = 3e^{i\frac{2\pi}{3}}$$

$$z_C = 2e^{i\frac{2\pi}{3}}$$

$$\{(B; 1), (C; -1), (D; 1)\}$$

$$O$$

$$D$$

$$-\frac{3}{2} - \frac{1 - 2\sqrt{3}}{2}i$$

$$-\frac{3}{2} + \frac{1 - 2\sqrt{3}}{2}i$$

$$1 + \frac{\sqrt{2}}{2}i$$

$$(z_B)^n$$

$$k \in \mathbb{N} \quad 8k + 1$$

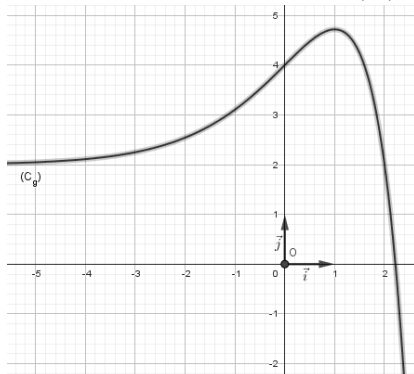
$$k \in \mathbb{N} \quad 4k + 1$$

$$k \in \mathbb{N} \quad 8k$$

التمرين الرابع: (07 نقاط)

$$g(x) = (2 - x)e^x + 2$$

$$g$$



$$\alpha$$

$$(O; \vec{i}, \vec{j})$$

$$g(x) = 0$$

$$2, 21 < \alpha < 2, 22$$

$$g(-x)$$

$$g(x)$$

$$f(x) = \frac{x^2}{1 + e^{-x}}$$

$$\mathbb{R}$$

$$f$$

$$(O; \vec{i}, \vec{j})$$

$$f$$

$$\lim_{x \rightarrow -\infty} f(x) \quad \lim_{x \rightarrow +\infty} f(x)$$

$$f$$

$$f'(x) = \frac{x \cdot g(-x)}{(1 + e^{-x})^2}$$

$$f$$

$$f(-\alpha)$$

$$f(-\alpha) = \alpha(\alpha - 2)$$

$$(I)$$

$$1, 1 < \beta < 1, 2$$

$$\beta$$

$$f(x) = 1$$

$$E$$

$$E(f(x)) = 0$$

$$\mathbb{R}$$

$$x \mapsto x^2$$

$$(\Gamma)$$

$$\lim_{x \rightarrow +\infty} [f(x) - x^2] = 0 \quad ($$

$$(\Gamma) \quad (C_f) \quad ($$

$$(f(-\alpha) \approx 0,48) \quad (\Gamma) \quad (C_f) \quad \mathbf{(5)}$$

$$f(x) = x^2 + \ln(m) : \quad m \quad \mathbf{(6)}$$

$$K \quad 0 \preceq f(x) \preceq x^2 : \quad x \quad \mathbf{(7)}$$

$$x = 0 ; x = 1 ; y = 0 \quad (C_f)$$

التمرين الأول: (05 نقاط)

2 1 0

10

4 2 1

3 2 1 0

:

:

"

" : A

" : B

" : C

2024

"2

1

X

(2

.

X

(

$$E(10X + b) = 2032 :$$

b

(

$$P\left(X = \frac{6}{5 - X}\right)$$

$$P(e^{X^2 - X} = 1)$$

(

التمرين الثاني: (04 نقاط)

$$p(z) = z^3 + 8 :$$

z

p(z) (I

: z ∈ ℂ

β α

p(-2)

(1

$$p(z) = (z + 2)(z^2 + \alpha z + \beta)$$

$$p(z) = 0 \quad \mathbb{C}$$

(2

$$(O; \vec{u}, \vec{v})$$

(II

$$z_C = -2 \quad z_B = \overline{z_A} \quad z_A = 1 + i\sqrt{3}$$

C B A

.

z_C z_B z_A :

(1

.

$$\frac{z_A - z_C}{z_B - z_C}$$

(

(2

$$z_B - z_C$$

(

$$ABC$$

(

$$ABC$$

O

(

$$(z_A)^n = (z_B)^n :$$

n

(3

$$z + 2 = ke^{i\frac{\pi}{6}} :$$

ℝ₊^{*}

k

z

(Γ)

(4

$$(\Gamma)$$

$$(\Gamma)$$

A

$$\begin{cases} \mathbf{u}_0 = 3 \ln 2 \\ \mathbf{u}_{n+1} = \ln \left(\frac{2}{7} e^{U_n} + 5 \right) \end{cases} : \quad \mathbb{N} \quad (U_n) \quad (I)$$

$$U_n \succ \ln 7 : n \quad (1)$$

$$e^{U_n} \succ e^{U_{n+1}} : \quad e^{U_n} - e^{U_{n+1}} = \frac{5}{7}(e^{U_n} - 7) : \quad (2)$$

$$\begin{aligned} & \lim_{n \rightarrow +\infty} U_n \quad (U_n) \quad (3) \\ V_n = e^{U_n} - 7 : \quad \mathbb{N} \quad (V_n) \quad (II) \\ & \quad \quad \quad (V_n) \quad (1) \end{aligned}$$

$$\begin{aligned} & \lim_{n \rightarrow +\infty} U_n \quad U_n = \ln \left(7 + \left(\frac{2}{7} \right)^n \right) : n \quad (2) \\ & \quad \quad \quad : \quad n \quad (3) \end{aligned}$$

$$S_n = 2024^{\ln(e^{U_0}-7)} + 2024^{\ln(e^{U_1}-7)} + 2024^{\ln(e^{U_2}-7)} + \dots + 2024^{\ln(e^{U_n}-7)}$$

$$f(x) = \frac{1-x^2}{x^2} + 2 \ln x : \quad]0; +\infty[\quad f \quad (I)$$

$$\lim_{x \rightarrow +\infty} f(x) \quad \lim_{x \rightarrow 0} f(x) \quad (1)$$

$$f \quad (2)$$

$$\begin{aligned} &]0; +\infty[\quad f(x) \quad x \quad f(1) \quad (3) \\ g(x) = x - 2 + \frac{1}{2}(\ln x)^2 + \frac{4 + 3 \ln x}{x} : \quad]0; +\infty[\quad g \quad (II) \\ & (O; \vec{i}, \vec{j}) \quad g \quad (C_g) \end{aligned}$$

$$\lim_{x \rightarrow +\infty} g(x) \quad (1)$$

$$g(x) = x - 2 + \frac{\ln x}{x} \left[3 + \frac{4}{\ln x} + \frac{1}{2} x \ln x \right] : x \in]0; +\infty[\quad (2)$$

$$\lim_{x \rightarrow 0} g(x) \quad ($$

$$g'(x) = \frac{1-x}{x^2} \ln \left(\frac{1}{x} \right) + \frac{1}{x^2} f \left(\frac{1}{x} \right) : x \in]0; +\infty[\quad (3)$$

$$g \quad ($$

$$(C_g) \quad (4)$$

$$(C_g) \quad (T) \quad ($$

$$0.2 \prec \alpha \prec 0.3 \quad \alpha \quad (C_g) \quad (5)$$

$$(C_g) \quad (6)$$